

IP Addressing & Subnetting

Question 1: List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Answer:

Class	First octet range	Default subnet mask	Purpose	Real-world example
A	1 - 126	255.0.0.0 (/8)	Very large networks (about 16 million hosts each)	Google Public DNS server (8.8.8.8); large ISP and corporate networks
B	128 - 191	255.255.0.0 (/16)	Medium to large networks (about 65,534 hosts each)	University or college campus network
C	192 - 223	255.255.255.0 (/24)	Small networks (254 hosts each)	Home Wi-Fi router and connected phones or laptops (192.168.1.x)
D	224 - 239	Not applicable	Multicast (one-to-many delivery)	IPTV / live video streaming; OSPF routers (224.0.0.5)
E	240 - 255	Not applicable	Reserved for experimental and research use	Experimental testing in research labs (not used on the public Internet)

Question 2: Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Answer:

Step 1: Convert the IP address and subnet mask to binary.

	Decimal	Binary
IP address	192.168.10.45	11000000.10101000.00001010.00101101
Subnet mask	255.255.255.0	11111111.11111111.11111111.00000000

Step 2: Network address = IP address AND subnet mask.

11000000.10101000.00001010.00101101 AND 11111111.11111111.11111111.00000000 =
11000000.10101000.00001010.00000000

Network address = 192.168.10.0

Step 3: Broadcast address = set all host bits to 1.

The mask has 24 network bits, so the last 8 bits are host bits:

11000000.10101000.00001010.**11111111**

Broadcast address = 192.168.10.255

Step 4: Range of valid host addresses.

First host = network address + 1 = **192.168.10.1**

Last host = broadcast address - 1 = **192.168.10.254**

Number of usable hosts = $2^8 - 2 = 254$

Network address	Broadcast address	Valid host range
192.168.10.0	192.168.10.255	192.168.10.1 to 192.168.10.254

Question 3: Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Answer:

Calculation: 4 subnets are needed. Borrowing n bits gives 2^n subnets, so $2^2 = 4$ means n = 2 bits.

New prefix = $24 + 2 =$ **/26**

Subnet mask = 11111111.11111111.11111111.11000000 = **255.255.255.192**

Block size = $256 - 192 = 64$ addresses per subnet

Usable hosts per subnet = $2^6 - 2 = 62$

Group	Network address	Subnet mask	Valid host range	Broadcast address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 - 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 - 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 - 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 - 10.0.0.254	10.0.0.255

Question 4: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Answer:

Step 1: Find the number of bits to borrow.

We need $2^n \geq 20$ subnets.

$2^4 = 16$ (not enough), $2^5 = 32$ (enough). So n = **5 bits**.

Step 2: Find the new subnet mask.

New prefix length = $16 + 5 =$ **/21**

Binary mask = 11111111.11111111.11111000.00000000

Third octet = $128 + 64 + 32 + 16 + 8 = 248$

New subnet mask = 255.255.248.0

Step 3: Find the hosts per subnet.

Host bits = $32 - 21 = 11$

Hosts per subnet = $2^{11} - 2 = 2048 - 2 =$ **2046 usable hosts**

Final answers:

Item	Value
New subnet mask	255.255.248.0 (/21)
Number of subnets created	32 (block size of 8 in the third octet)
Hosts available per subnet	2046
Example subnets	172.16.0.0/21, 172.16.8.0/21, 172.16.16.0/21, ...

Question 5: A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Answer:

Hi! The problem is with the IP address itself. An IPv4 address has four octets, and each octet is an 8-bit number, so it can only be between 0 and 255. In 192.168.1.300 the last octet is 300, which is more than 255, so it is not a valid IP address and the device cannot use it to join the network.

Since the address starts with 192.168.1, the device is on the 192.168.1.0/24 network (subnet mask 255.255.255.0). A valid host address in this subnet must be between 192.168.1.1 and 192.168.1.254, for example **192.168.1.30** or **192.168.1.100**. Do not use 192.168.1.0 (network address) or 192.168.1.255 (broadcast address), and do not use the address already taken by the router (usually 192.168.1.1). The easiest fix is to set the device to obtain an IP address automatically (DHCP).